Lecture 7:-

Rules of Inference

→ If you have a current password (p) then you can log into the network (q). $p \rightarrow q$

→ You have a current password (p). p

therefore

→ You can log into the network (q). q.

Premises.

Hypothesis $\left\{ \begin{array}{ll} p \rightarrow q & - p1 \\ p & - p2 \end{array} \right.$

Conclusion $\therefore q - c$

Argument.

General form -

Premises.

$$\left.\begin{array}{l} p1 \\ p2 \\ \vdots \\ pN \end{array}\right\} \text{Hypothesis}$$

$\therefore C$ - Conclusion.

Argument.

Checking Validity of Argument.

$(P1 \wedge P2 \wedge P3 \wedge \ldots \wedge PN) \rightarrow C$

Result in tautology

Yes → Valid.

No → fallacy.

$p \to q$ — p1

p — p2

$\therefore q$ — c

$$((p \to q) \land p) \to q$$

p	q	$p \to q$	$(p \to q) \land p$	$((p \to q) \land p) \to q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

X. L3 !

P61.

1) p, $p \to q$, $\therefore q$ — Modus Ponens.

2) $\neg q$, $p \to q$, $\therefore \neg p$ — Modus Tollens.

3) $p \to q$, $q \to r$, $\therefore p \to r$ — Hypothetical Syllogism

4) $p \lor q$, $\neg p$, $\therefore q$. — Disjunctive Syllogism.

5) p, $\therefore p \lor q$ — Addition.

6) $p \land q$, $\therefore p$ — Simplification.

7) p, q, $\therefore p \land q$ — Conjunction.

8) $p \lor q$, $\neg q \lor r$, $\therefore p \lor r$. — Resolution.

Ex6 :- P62 "It is not ($\neg$) sunny (p) and ($\land$) it is colder (q)" $\neg p \land q$

Ex6 :-
P62

"It is (not) Sunny (and) it is Colder" $\neg p \wedge q$
(p = Sunny, q = it is Colder)

"we go swimming (only if) it is Sunny" $r \to p$
(r = go swimming, p = it is Sunny)

"(If) we do (not) go swimming (Than) we take a trip" $\neg r \to s$
(r = we go swimming, s = we take a trip)

"(If) we take a trip (Than) we will be home by Sunset" $s \to t$
(s = we take a trip, t = will be home)

Leads to Conclusion.

"We will be home by Sunset". t.
(t = will be home)

$\neg p \wedge q$ — P1 ✓
$r \to p$ — P2 ✓
$\neg r \to s$ — P3 ✓
$s \to t$ — P4
$\therefore t$. — C

from P1 $\neg p$ — (5) By Simplification
from (P2,5) $\neg r$ — (6) By MT
" (P3,6) s — (7) By MP.
" (P4,7) t — (8) " " .

which is Conclusion.
therefore Argument is Valid.

1) p, $p \to q$, $\therefore q$. — Modus Ponens.

2) $\neg q$, $p \to q$, $\therefore \neg p$ — Modus Tollens.

3) $p \to q$, $q \to r$, $\therefore p \to r$ — Hypothetical Syllogism

4) $p \vee q$, $\neg p$, $\therefore q$. — Disjunctive Syllogism.

5) p, $\therefore p \vee q$ — Addition.

6) $p \wedge q$, $\therefore p$ — Simplification.

7) p, q, $\therefore p \wedge q$ — Conjunction.

8) $p \vee q$, $\neg q \vee r$, $\therefore p \vee r$. — Resolution

Ex7 :-
P62-63

$p \to q$ — P1 ✓.
$\neg p \to r$ — P2
$r \to s$ — P3
$\therefore \neg q \to s$. — C.

from (P1) $\neg q \to \neg p$ — (4) By

1) p, $p \to q$, $\therefore q$. — Modus Ponens.

2) $\neg q$, $p \to q$, — Modus Tollens

5) p, $\therefore p \vee q$ — Addition.

6) $p \wedge q$, $\therefore p$ — Simplification.

$\therefore \neg q \to s$ — C.

from (P1) $\neg q \to \neg p$ — (4) By Contrapositive.

" (P2,4) $\neg q \to r$ — (5) By HS.

" (P3,5) $\neg q \to s$ — (6) " ".

which is conclusion.

$\therefore$ Argument Valid.

2) $\neg q$, $p \to q$, $\therefore \neg p$ — Modus Tollens.

3) $p \to q$, $q \to r$, $\therefore p \to r$ — Hypothetical Syllogism

4) $p \vee q$, $\neg p$, $\therefore q$. — Disjunctive Syllogism.

6) $p \wedge q$, $\therefore p$ — Simplification.

7) p, q, $\therefore p \wedge q$ — Conjunction.

8) $p \vee q$, $\neg q \vee r$, $\therefore p \vee r$. — Resolution

Issue 1:- Alot of memorization. Alot of Hit-Trail.

Ex 8:-

P63

$T \to M \vee E$ — P1

$S \to \neg E$ — P2

$T \wedge S$ — P3

$\therefore M$.

from (P3) T — (4) By S

" " S — (5) " ".

" (P1,4) $M \vee E$ — (6) By MP.

" (P2,5) $\neg E$ — (7) " ".

" (6,7) M — (8) By DS.

Which is Conclusion.

$\therefore$ Valid.

1) p, $p \to q$, $\therefore q$. — Modus Ponens.

2) $\neg q$, $p \to q$, $\therefore \neg p$ — Modus Tollens.

3) $p \to q$, $q \to r$, $\therefore p \to r$ — Hypothetical Syllogism

4) $p \vee q$, $\neg p$, $\therefore q$. — Disjunctive Syllogism.

5) p, $\therefore p \vee q$ — Addition.

6) $p \wedge q$, $\therefore p$ — Simplification.

7) p, q, $\therefore p \wedge q$ — Conjunction.

8) $p \vee q$, $\neg q \vee r$, $\therefore p \vee r$. — Resolution

Issue 2:- Sequencing.

"Principle of Resolution."